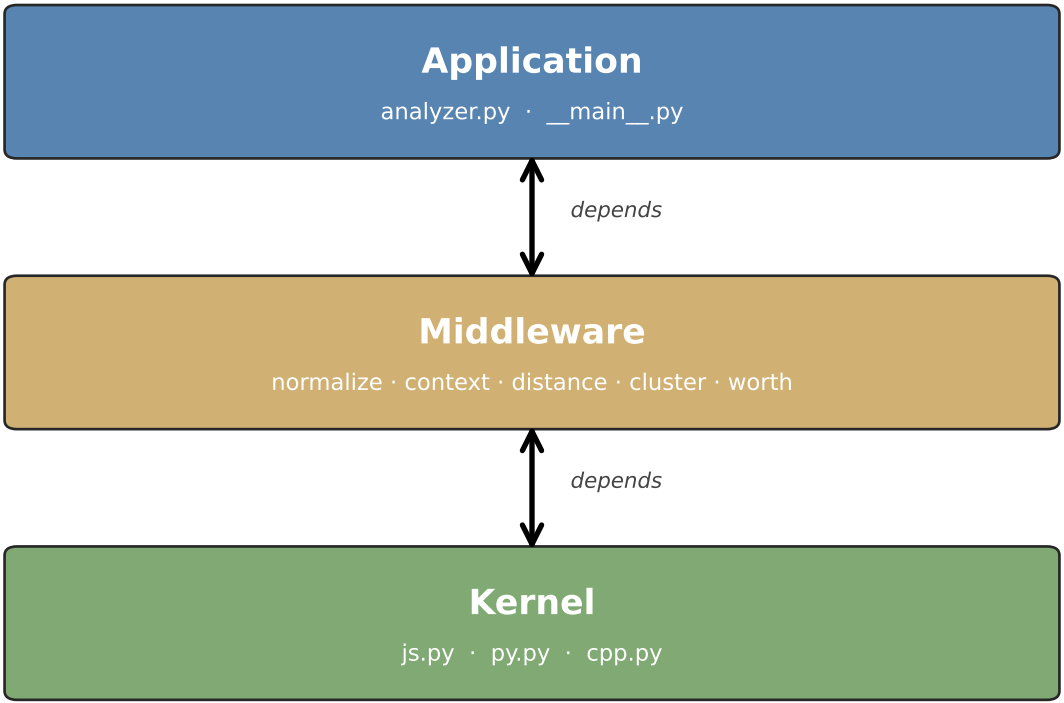DAG: application → middleware → kernel